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Research Article

Constructing Efficient Identity-Based Signatures on Lattices

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In this work, we explore the recent developments related to lattice-based signature and preimage sampling, and specify a compact identity-based signature (IBS) on an ideal lattice for practical use. Specifically, we first propose an ellipsoid version of the G + G signature scheme (Asiacrypt 2023) that achieves slightly better signature size and higher security. Then, by adapting a specific preimage sampling algorithm to the modified G + G signature, we obtain an efficient IBS scheme. In addition, we prove its security in the quantum random oracle model (QROM), following the paradigm introduced by Zhangdry (Crypto 2012). Finally, a complete specification of the IBS, featuring three distinct parameter sets, is accompanied by a proof-of-concept implementation. We believe that the combination of the preimage sampling with the Fiat–Shamir transformation holds potential for application in the other advanced digital signature schemes.

1. Introduction

Digital signature is of particular importance for authentication between users, especially in open network systems. However, the traditional authentication requires dedicated infrastructure and onerous certificate management. As a result, it is less suitable for scenarios where constrained devices are involved.

1.1. Identity-Based Signature (IBS). The concept of IBS, which was pioneered by Shamir [1], has emerged as a promising substitute for the traditional signature schemes, since the public key of every user is computed based solely on its identity information, and there exists a trusted authority that extracts the associated private key by using a master secret key. In such a manner, the key generation and distribution procedures are greatly simplified, making IBS an important tool for lightweight authentication. The early constructions of IBS, mainly based on pairings [2–5], are efficient in terms of both time and space, but vulnerable to quantum attacks due to Shor's algorithm [6].

1.2. Lattice-Based IBS. Lattice-based cryptography is a desirable post-quantum alternative to traditional cryptography. In the past decade, much work has been put into improving its practical efficiency. Specifically, the efficiency of basic cryptographic operations, namely encryption and signing, is competitive with established classical methods. The first four post-quantum algorithms selected for standardization by NIST in July 2022 were dominated by lattice-based constructions, including Kyber [7] for encryption/key encapsulation, alongside Dilithium [8] and Falcon [9] for digital signatures.

As for lattice-based IBS, in 2010 Rückert [10] presented the first two lattice-based (hierarchical) IBS constructions in the random oracle model (ROM) and standard model, respectively, followed by efficiency improvements in [11, 12]. Technically, these schemes follow the "trapdoor delegation [13–15] + GPV signature [16]" paradigm, where the former generates a short basis that serves as the secret key in the later procedure. In theory, these schemes are elegant and exemplify the versatility of the preimage sampling [16], a core component that serves as the foundation of the trapdoor delegation and GPV signature, as well as many other lattice-based advanced primitives. However, these IBSs have poor performance in time and space, due to the trapdoor delegation procedure, which is not only time-consuming but also increases parameters

dramatically, leading to a large key size and signature size. Recently, Sageloli et al. [17] followed this paradigm and provided lattice-based IBS schemes with tight security against adaptive adversaries in the quantum ROM (QROM), by improving on [18] in several aspects. Due to the costly trapdoor delegation, however, its parameter estimations indicate that the signature is a few megabytes in size, at least three orders of magnitude larger than those of ordinary signature schemes, for example, Dilithium [8]. Even worse, a user's private key can be several hundred megabytes to several gigabytes in size.

Another paradigm, that is, "preimage sampling [16] + rejection sampling [19]," was first proposed by Tian and Huang [20] to avoid the costly trapdoor delegation. From a high-level technical point of view, it can be seen as an identity-based version of Lyubashevsky's signature scheme. IBS in this paradigm is more efficient in both time and space, because it no longer requires trapdoor delegation, and the rejection sampling procedure used for signing messages is usually simpler than GPV. Indeed, Yang et al. [21] instantiate this paradigm by "MP12 sampler [22] + Lyubashevsky signature [19]," whose performance outperforms the previous construction [20]. However, their implementation mainly aimed at demonstrating the time efficiency of lattice-based IBS, without a concrete security analysis. Besides, due to the use of the MP12 sampler, the instantiation suffers from a huge public key size.

1.3. Challenges for Compact Lattice-Based IBS in Practice. As mentioned before, "preimage sampling [16] + rejection sampling [19]" gives more efficient IBS, in which the preimage sampling generates users' secret keys. However, the efficiency here is only in an asymptotic sense, and the commonly used preimage sampling algorithms [16, 22, 23] usually result in inefficient instantiations, especially in the master public key and signature sizes. In more detail, a fat matrix $\mathbf{A} \in \mathbb{Z}_Q^{n \times m}$ and its corresponding trapdoor $\mathbf{T}$ denote the master public key and the master secret key, respectively. For any identity $id \in \{0, 1\}^*$, the trusted authority derives a user-specific sk_{id} from $\mathbf{T}$. This is achieved by first hashing id to a vector $\mathbf{u} \in \mathbb{Z}_Q^n$, then generating a short vector $\mathbf{x}$ sampled from a discrete Gaussian distribution $D_{\mathbb{Z}^m,\sigma}$ under the constraint that $\mathbf{Ax} = \mathbf{u} \bmod Q$, which serves as the user's private key. Given a message, the signing algorithm mainly produces a masking vector $\mathbf{y} \in \mathbb{Z}^m$ with a specific parameter such that it is large enough that the eventual signature does not reveal the secret key (i.e., the signing algorithm is zero-knowledge), yet small enough so that the signature is not easily forged.

Recall that in Lyubashevsky's signature scheme [19], each entry of the masked vector $\mathbf{y}$ has magnitude $\widetilde{\Theta}(\sqrt{m})$, even the secret key $\mathbf{x} \in \mathbb{Z}^m$ is sampled from a distribution with a very small deviation. This explains why Dilithium is comparable in speed to traditional signature algorithms, but has nearly 40 times the signature size. The situation becomes worse if its identity-based version is considered, that is, the secret key is extracted by a preimage sampling algorithm, because in this case the standard deviation of the distribution that the secret key follows is increased from a constant to $\widetilde{\Omega}(\sqrt{m})$, which leads to a drastic increase in the parameter, and consequently, the master public key and the signatures become quite large.

Table 1: Summarized performance of SRGG-IBS.

Target security	100	180	256
$\|mpk\|$ (in bytes)	3104	4640	6176
$\|sig\|$ (in bytes)	12,088	17,736	23,480
Extract (in cycles)	5,004,676	5,490,761	9,556,676
Sign (in cycles)	5,203,248	7,007,089	11,521,424
Security C/Q	103/93	171/155	242/219

Although there is still no concrete implementation of the "MP12 sampler + Lyubashevsky signature" paradigm so far,[1] according to our estimation, the master public key size would be more than a megabyte under NIST-1 security level, because usually the size of IBS is larger than that of identity-based encryption (IBE) [24] at the same security level. In summary, preimage sampling and digital signature together determine the efficiency of IBS, and it is challenging to design and implement an efficient IBS for practical use.

1.4. Contributions. In this work, we mainly aim at specifying a compact IBS on an ideal lattice for practical use by exploring the recent developments related to lattice-based signatures. Similar to the "preimage sampling + rejection sampling" paradigm, our work combines the preimage sampling that is more suitable for IBE [25], with an ellipsoid version of Fiat–Shamir heuristic-based G + G signature [26], toward improving the concrete efficiency. We summarize our principal contributions as follows:

- *Ellipsoid G + G*: The G + G signature scheme [26] generally follows the Fiat–Shamir heuristic but makes use of the Gaussian convolution technique [22, 23] in terms of detail, which explains the name "Gaussian plus Gaussian." Notice that the G + G signature avoids the rejection step [19] due to the preimage sampling. Our first contribution is to incorporate the ellipsoid Gaussian [27] into the G + G signature. By analyzing the optimal parameters, we obtain an improved signature scheme that has a slightly more compact signature and higher security than the G + G signature [26].
- *Efficient construction of IBS*: Having the ellipsoid G + G signature, we adapt the IBE-friendly preimage sampling [25] to this setting, and obtain an efficient IBS construction, which may be viewed as an identity-based adaptation of the G + G signature. Then, we prove its EUF-ID-CMA security in the QROM, which reduces its security to the security of the low-level induced G + G signature scheme.
- *Instantiation and implementation*: Finally, we give a specification of our IBS with three parameter sets on an ideal lattice, named SRGG-IBS. A proof-of-concept implementation was developed to assess its practicality, with the corresponding performance metrics detailed in Table 1.

1.5. Other Post-Quantum IBS. Currently, in addition to lattice-based IBS, there are several post-quantum IBS constructions

relying on various hardness assumptions. Debnath et al. [28] proposed a multivariate-based IBS that utilizes a multivariate signature scheme along with a 5-pass identification scheme in the ROM. When instantiated with Rainbow, the scheme achieves moderate efficiency. Besides, Chen et al. [29] put forward a tightly secure IBS scheme based on the supersingular isogeny problems in the ROM, but without a concrete instantiation.

1.6. Rode Map. We commence with preliminary materials in Section 2, subsequently introducing our enhanced ellipsoid G + G in Section 3. Our IBS construction is then delineated in Section 4. Section 5 delineates three distinct parameter sets and assesses the implementation's performance. The concluding remarks are provided in Section 6.

2. Preliminary

Let $\mathbb{R}$ and $\mathbb{Z}$ represent the sets of real numbers and integers, respectively. For a real-valued function f defined on a countable set S, we define $f(S) = \sum_{x\in S} f(x)$, provided that this series converges absolutely.

2.1. Linear Algebra and Lattices. A vector is represented in bold lowercase, for example, $\mathbf{x} = (x_1, \ldots, x_n)^\top$, and it is treated as a column vector. The concatenation of two vectors $\mathbf{x}_1$ and $\mathbf{x}_2$ is written as $(\mathbf{x}_1, \mathbf{x}_2)$. We define the inner product of $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ as $\langle \mathbf{x}, \mathbf{y}\rangle$, and the ℓ_2 norm of $\mathbf{x}$ as $|\mathbf{x}| = \sqrt{\langle \mathbf{x}, \mathbf{x}\rangle}$. Matrices are denoted in bold uppercase, for example, $\mathbf{A} = [\mathbf{a}_1| \cdots |\mathbf{a}_n]$, where $\mathbf{a}_i$ represents the i-th column of $\mathbf{A}$. The tensor product is denoted by $\otimes$, and the block diagonal concatenation of matrices $\mathbf{A}$ and $\mathbf{B}$ by $\mathbf{A} \oplus \mathbf{B}$. The largest singular value of $\mathbf{A}$ is defined as $s_1(\mathbf{A}) = \max_{\mathbf{x}\neq \mathbf{0}} \frac{\|\mathbf{Ax}\|}{\|\mathbf{x}\|}$, and the transpose of $\mathbf{A}$ is written as $\mathbf{A}^\top$.

A symmetric matrix $\Sigma \in \mathbb{R}^{m\times m}$ is said to be positive definite, denoted $\Sigma > 0$, if $\mathbf{x}^\top \Sigma \mathbf{x} > 0$ for all nonzero $\mathbf{x} \in \mathbb{R}^m$. We write $\Sigma_1 > \Sigma_2$ when $\Sigma_1 - \Sigma_2 > 0$, and for a scalar s, we write $\Sigma > s$ if $\Sigma - s \cdot \mathbf{I} > 0$.

Given a matrix $\mathbf{B} = [\mathbf{b}_1| \cdots |\mathbf{b}_n] \in \mathbb{R}^{m\times n}$ whose columns $\mathbf{b}_i$ are linearly independent, the lattice produced by $\mathbf{B}$ is defined as $\Lambda(\mathbf{B}) = \{\mathbf{Bz} | \mathbf{z} \in \mathbb{Z}^n\}$. The dual lattice of Λ is given by $\Lambda^* = \{\mathbf{y} \in \mathrm{span}(\Lambda) | \langle \mathbf{x}, \mathbf{y}\rangle \in \mathbb{Z}, \forall \mathbf{x} \in \Lambda\}$. In lattice-based cryptography, the q-ary lattice plays a notably important role and is constructed from a matrix $\mathbf{A} \in \mathbb{Z}_q^{n\times m}$ as follows:

$$\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m : \mathbf{Ax} = \mathbf{0} \mod q\}.$$

For any $\mathbf{u} \in \mathbb{Z}_q^n$ and $\mathbf{x} \in \mathbb{Z}^m$ such that $\mathbf{A} \cdot \mathbf{x} = \mathbf{u} \mod q$, the "shifted lattice" is the set as follows:

$$\Lambda_{\mathbf{u}}^\perp(\mathbf{A}) = \{\mathbf{z} \in \mathbb{Z}^m : \mathbf{A} \cdot \mathbf{z} = \mathbf{u} \mod q\} = \Lambda_q^\perp(\mathbf{A}) + \mathbf{x}.$$

2.2. Gaussians. The Gaussian function $\rho : \mathbb{R}^m \to (0, 1]$ is defined as $\rho(\mathbf{x}) = \exp(-\pi \cdot \langle \mathbf{x}, \mathbf{x}\rangle)$. For any positive definite Σ and $\mathbf{c} \in \mathrm{span}(\Sigma)$, we define the following:

$$\rho_{\Sigma,\mathbf{c}}(\mathbf{x}) = \exp(-\pi \cdot (\mathbf{x} - \mathbf{c})^t \Sigma^{-1} (\mathbf{x} - \mathbf{c})).$$

The normalization of $\rho_{\Sigma,\mathbf{c}}$ yields the continuous Gaussian distribution $D_{\Sigma,\mathbf{c}}$. Restricting the support of the distribution to the lattice Λ, we get the following:

$$D_{\Lambda,\Sigma,\mathbf{c}}(\mathbf{x}) = \frac{\rho_{\Sigma,\mathbf{c}}(\mathbf{x})}{\rho_{\Sigma,\mathbf{c}}(\Lambda)}.$$

Let $\eta_\varepsilon(\Lambda) = \min\{s > 0 | \rho(s \cdot \Lambda^*) \leq 1 + \varepsilon\}$ be the smoothing parameter with respect to a lattice Λ and $\varepsilon \in (0, 1)$.

2.3. Cyclotomics. Let $\mathbb{Z}_m^*$ be the set of the $d = \varphi(m)$ integers that are invertible modulo m. Let ζ_m be a primitive m-th primitive root of unity, and let the m-th cyclotomic polynomial be defined as $\Phi_m(X) = \sum_{i\in\mathbb{Z}_m^*} (X - \zeta_m^i) \in \mathbb{Z}[X]$. We define the cyclotomic ring of conductor m as $\mathscr{R} = \mathbb{Z}[\zeta_m] \simeq \mathbb{Z}[X]/(\Phi_m(X))$, and the corresponding cyclotomic field as $\mathscr{K} = \mathbb{Q}[\zeta_m] \simeq \mathbb{Q}[X]/(\Phi_m(X))$. Let $\mathscr{K}_\mathbb{R} = \mathscr{K} \otimes \mathbb{R} = \mathbb{R}[\zeta_m] \simeq \mathbb{R}[X]/(\Phi_m(X))$. In this article, we focus on the case of a power-of-two conductor, that is, $m = 2^t$ for some $t \in \mathbb{Z}$.

Any element $f \in \mathscr{K}$ admits a unique representation of the form $f = \sum_{i=0}^{d-1} f_i \zeta_m^i$ with $f_i \in \mathbb{Q}$. We call $(f)_c = (f_0, \ldots, f_{d-1}) \in \mathbb{Q}^d$ the coefficient embedding (or coefficient vector) of f. The element of $f \in \mathscr{K}$ also admits a matrix representation $\mathscr{M}(f) = \left[(f)_c | (\zeta_m f)_c | (\zeta_m^{d-1} f)_c\right] \in \mathbb{Q}^{d\times d}$.

There are d embeddings of $\mathscr{K}$ fixing over $\mathbb{Q}$. Concretely, for $i \in \mathbb{Z}_m^*$, the embedding σ_i is given by the relation $\sigma_i(\zeta_m) = \zeta_m^i$. These d embeddings constitute the singular values of $\mathscr{M}(f)$. The canonical embedding for an element $f \in \mathscr{K}$ is defined as $\sigma(f) = (\sigma_i(f))_{i\in\mathbb{Z}_m^*}$. The conjugate of f is represented by f^*.

2.4. IBS

Definition 1 (IBS). Let $\mathscr{ID}$ be the identity space. An IBS scheme consists of polynomial-time algorithms (Setup, Extract, Sign, Vrfy).

- Setup(1^λ): given the security parameter 1^λ, the algorithm generates a master public key and master secret key pair (mpk, msk).
- Extract(mpk, msk, id): using the master public key mpk, master secret key msk, and a user identity $id \in \mathscr{ID}$, the algorithm generates a secret key sk_{id} for that identity.
- Sign(mpk, id, sk_{id}, μ): on input mpk, id, sk_{id} and a message μ, output a signature sig.
- Vrfy(mpk, id, μ, sig): given mpk, id, μ and sig, output 1 (accept) or 0 (reject).

Definition 2 (Correctness). An IBS scheme (Setup, Extract, Sign, Vrfy) with identity space $\mathscr{ID}$ is correct, if for every $(mpk, msk) \leftarrow$ Setup(1^λ), every identity $id \in \mathscr{ID}$ and message $\mu \in \{0, 1\}^*$, it holds that Vrfy$(mpk, id, \mu, sig) = 1$, where $sk_{id} \leftarrow$ Extract(mpk, msk, id) and $sig \leftarrow$ Sign(mpk, id, sk_{id}, μ).

The security notion for IBS adopted in this article is the existential unforgeability against chosen identity and chosen

message attacks (EUF-ID-CMA security), formalized through an interactive game between a challenger and an adversary.

Definition 3 (EUF-ID-CMA). An IBS scheme (Setup, Extract, Sign, Vrfy) is EUF-ID-CMA secure, if for any PPT adversary $\mathcal{A}$, its advantage as follows:

$$\mathsf{Adv}^{\text{euf-id-cma}}_{\mathsf{IBS},\mathcal{A}}(\lambda) = Pr\left[\mathsf{Exp}^{\text{euf-id-cma}}_{\mathsf{IBS},\mathcal{A}}(\lambda) = 1\right] = \mathsf{negl}(\lambda),$$

where $\mathsf{Exp}^{\text{euf-id-cma}}_{\mathsf{IBS},\mathcal{A}}(\lambda)$ is defined by the following.

1. Setup: The challenger generates $(mpk, msk) \leftarrow \mathsf{Setup}(1^\lambda)$ and initializes two empty lists $\mathcal{L}_{sk_{id}}$ and $\mathcal{L}_\mu$. It gives mpk to the adversary $\mathcal{A}$.
2. Queries: The adversary $\mathcal{A}$ may query the following oracles:
   - Key extraction oracle $\mathcal{O}_{\text{extract}}$: On input $id \in \mathcal{ID}$, $\mathcal{O}_{\text{extract}}$ checks whether $(id, sk_{id}) \in \mathcal{L}_{sk_{id}}$ for some sk_{id}. If such sk_{id} exists, it returns sk_{id}. Otherwise, it returns $sk_{id} \leftarrow \mathsf{Extract}(mpk, msk, id)$ and sets $\mathcal{L}_{sk_{id}} \leftarrow \mathcal{L}_{sk_{id}} \cup \{id, sk_{id}\}$.
   - Signing oracle $\mathcal{O}_{\text{sign}}$: On input (id, μ), $\mathcal{O}_{\text{sign}}$ sets $\mathcal{L}_\mu \leftarrow \mathcal{L}_\mu \cup \{(id, \mu)\}$, and checks whether $(id, sk_{id}) \in \mathcal{L}_{sk_{id}}$. If such sk_{id} exists, it returns $sig \leftarrow \mathsf{Sign}(mpk, id, sk_{id}, \mu)$. Otherwise, it computes $sk_{id} \leftarrow \mathsf{Extract}(mpk, msk, id)$, updates the secret key lists $\mathcal{L}_{sk_{id}} \leftarrow \mathcal{L}_{sk_{id}} \cup \{id, sk_{id}\}$ and finally returns $sig \leftarrow \mathsf{Sign}(mpk, id, sk_{id}, \mu)$. For both situations, the tuple (id, μ) is recorded by the challenger in $\mathcal{L}_\mu$.
3. Output: $\mathcal{A}$ outputs a forgery (id^*, μ^*, sig^*). The challenger outputs 1 if $\mathcal{A}$ never queries id^*'s secret key sk_{id^*}, $(id^*, \mu^*) \notin \mathcal{L}_\mu$ and $\mathsf{Vrfy}(mpk, id^*, \mu^*, sig^*) = 1$; otherwise, it outputs 0.

3. Ellipsoid G + G

This section constructs an ellipsoid G + G identification protocol by incorporating ellipsoid Gaussian [27] into G + G [26], and then transforms it into a signature using the Fiat–Shamir heuristic.

3.1. Ellipsoid G + G Identification Protocol. In the following, we define parameters where $m \geq l > 0, k > m + l$. Let $\mathbf{J} \in \mathbb{Z}^{m \times l}$ be the matrix with its first m diagonal elements set to 1 and all others 0. Let χ be a distribution over $\mathbb{Z}$ and denote $\mathcal{C} \subset \mathbb{Z}_2^l$ the challenge space. Additionally, an odd modulus q and an acceptance bound γ are required.

The G + G protocol works as follows: First sample a public key $\mathbf{A} \in \mathbb{Z}_{2q}^{m \times k}$ and a secret key $\mathbf{T} \in \mathbb{Z}^{k \times l}$ such that $\mathbf{A} \cdot \mathbf{T} = q \cdot \mathbf{J} \bmod 2q$. To authenticate itself, the prover samples a perturbation vector $\mathbf{y}$ and sends $\mathbf{w} = \mathbf{A} \cdot \mathbf{y} \bmod 2q$ to the verifier as the commitment. The verifier then draws a challenge vector $\mathbf{c}$ uniformly at random from the challenge space and sends it back to the prover. Finally, the prover samples a proper Gaussian vector $\mathbf{k}$ and returns $\mathbf{z} = \mathbf{y} + \mathbf{S} \cdot (2\mathbf{k} + \mathbf{c})$. Refer to [26] for more details.

Let $\sigma, s_1, s_2 > 0$. The novelty of the ellipsoid G + G is reflected in the covariance matrix $\Sigma_{\mathbf{p}} = s_1^2 \cdot \mathbf{I}_{k-l} \oplus s_2^2 \cdot \mathbf{I}_l - \sigma^2 \mathbf{T} \cdot \mathbf{T}^t$, which causes $\mathbf{z}$ to follow an ellipsoid Gaussian distribution characterized by s_1 and s_2. For correctness, the covariance matrix $\Sigma_{\mathbf{p}}$ should be positive definite.

3.2. Ellipsoid G + G Signature Scheme. Applying the Fiat–Shamir heuristic to this identification protocol yields a lattice-based signature $\mathsf{FS}[\text{ellipsoid}\,\mathsf{G} + \mathsf{G}, H]$.

Theorem 1 ([26], adapted). *Let $m \geq l > 0, k > m + l, \varepsilon \in (0, 1/2], \sigma \geq \sqrt{2\ln(l - 1 + 2l/\varepsilon)/\pi}$ and large enough s_1, s_2 such that $\Sigma_{\mathbf{p}}$ is positive definite. Let γ and ε_c be such that $\Pr_{\mathbf{z} \leftarrow D_{\mathbb{Z}^k, s}}[\|\mathbf{z}\| \geq \gamma] \leq \varepsilon_c/3$. Let $q > \max(2\gamma, s_1 \cdot \eta_\varepsilon(\mathbb{Z}^m))$ be an odd modulus. Then the signature scheme $\mathsf{FS}[\text{ellipsoid}\,\mathsf{G} + \mathsf{G}, H]$ is ε_c-correct and EUF-CMA-secure in the (Q)ROM.*

The proof of the above theorem requires some necessary properties of the underlying identification protocol, including completeness, commitment recoverability, honest-verifier zero-knowledge, commitment min-entropy, and lossy special soundness. The proof of these properties is omitted, as they can be derived directly from those in [26].

3.3. On Choosing the Parameters. All we need to do is to choose proper parameters s_1, s_2 such that the ellipsoid G + G identification protocol demonstrates superior performance compared to the original version. In more detail, by our definition, the perturbation covariance matrix is as follows:

$$\Sigma_{\mathbf{p}} = \begin{bmatrix} s_1^2 \cdot \mathbf{I}_{k-l} - \sigma^2 \cdot \mathbf{R}\mathbf{R}^t & -\sigma^2 \cdot \mathbf{R} \\ -\sigma^2 \cdot \mathbf{R}^t & (s_2^2 - \sigma^2) \cdot \mathbf{I}_l \end{bmatrix},$$

which should be positive definite for correctness. Besides, observe that the ℓ_2 norm of the final vector $\mathbf{z}$, which determines the concrete security, is upper-bounded by $\sqrt{s_1^2 \cdot (k - l) + s_2^2 \cdot l}$ with overwhilming probability. According to the analysis in [27], to make $\|\mathbf{z}\|$ as small as possible under the condition that $\Sigma_{\mathbf{p}}$ is positive definite, we can let the following:

$$s_1^2 = \left(1 + \frac{1}{s_1(\mathbf{R})}\sqrt{\frac{l}{k - l} \cdot s_1(\mathbf{R})}\right) \cdot \sigma^2 \cdot (s_1(\mathbf{R}))^2,$$
$$s_2^2 = \left(1 + \sqrt{\frac{k - l}{l} \cdot s_1(\mathbf{R})}\right) \cdot \sigma^2.$$

Note that s_1 will be slightly larger than the Gaussian parameter s in G + G, but s_2 will be drastically reduced. In this way, the preimage norm and the storage size are decreased.

3.4. Performance Comparison. Let $m > 0$, $k > m + 1$, and fix $l = 1$. Define the vector $\mathbf{j} = (\zeta^*, 0, \ldots, 0) \in \mathcal{R}^m$, where $\zeta =$

Table 2: Comparison between the original G + G [26] with our ellipsoid version.

Parameters	NIST-I level		NIST-III level		NIST-V level	
	[26]	Ours	[26]	Ours	[26]	Ours
$(m, k-m)$	(2,3)	(2,3)	(3,4)	(3,4)	(4,5)	(4,5)
σ	80.8	(83.3,5.4)	92.9	(95.1,6.1)	103.1	(105.1,6.1)
γ	3971.6.2	2694.7	3971.6.2	3766.8	4998.0	4806.9
Key recov. (C/Q)	112/102	112/102	183/166	183/166	256/233	256/233
Forg. security (C/Q)	113/102	**116/105**	179/163	**182/166**	246/223	**249/226**
Sig. size (bytes)	1360	**1232**	1932	**1808**	2520	**2428**
Pub. key size (bytes)	736	736	1088	1088	1440	1440

Note: The bold values demonstrate the areas where our solution exhibits superior performance compared to Devevey et al. [26].

$1 + X^{n/2}$ and $\zeta^* = 1 - X^{n/2}$ are such that $\zeta^*\zeta = 2 \bmod (1 + X^n)$. Table 2 shows a comparison of the performance *on the module lattice*. We fix the polynomial degree $n = 256$, modulus $q = 2047$, and the secret distribution $U(y \in \mathcal{R} \| \|y\|_\infty \le 1)$.

The resulting signature sizes are 3.6%–9.41% smaller than those in [26]. Besides, the ellipsoid version has several bits improvements in forgery security.

4. Specification of an Efficient IBS

This section gives a specification of our efficient IBS scheme, named SRGG-IBS. The parameters and notations are summarized in Table 3. For convenience, each element $f \in \mathcal{K}$ is used to represent its matrix form $\mathcal{M}(f)$. Note that the setup and extraction procedures follow from [25] in essence, except that we generalize the ring description to the module version.

4.1. Setup. SRGG-IBS uses the module LWE-style key pair. Its master public key is essentially given by $(\widehat{\mathbf{A}}, \mathbf{B} = \mathbf{F} - [\mathbf{I}_d, \widehat{\mathbf{A}}] \cdot \mathbf{R} \bmod Q)$, where $\widehat{\mathbf{A}} \sim U(\mathcal{R}_Q^{d \times d})$ and $\mathbf{F}$ is the gadget matrix. Refer to Algorithm 1 for its formal description. For compactness, an ideal extendable-output function is employed to produce the pseudorandom matrix $\widehat{\mathbf{A}}$, and $\mathsf{seed}_{\widehat{\mathbf{A}}}$ is stored as part of the *msk*.

For correctness, the covariance matrix $\Sigma_{\mathbf{p}}$ should be positive definite. By the properties of the Schur complement, checking the positive definiteness of $\Sigma_{\mathbf{p}}$ boils down to that of field elements in $\mathcal{K}_{\mathbb{R}}$, which can be realized by simply checking the positiveness of their canonical embedding.

4.2. Extract Users' Secret Keys. Upon receiving the master secret key *msk* and a user identity $id \in \mathcal{ID}$, the key extraction procedure detailed in Algorithm 2 generates an approximate preimage $\mathbf{y} \in \mathcal{R}^{(2+w)d}$ such that $\mathbf{A} \cdot \mathbf{y} \approx \mathsf{H}(id) \bmod Q$. This process is divided into offline and online phases, adopting the methodology introduced in [22, 23]. During the offline phase, a perturbation vector $\mathbf{p} \sim D_{\mathbb{Z}^{(2+w)d \cdot n}, \Sigma_{\mathbf{p}}}$ is sampled, a task that can be performed in advance. Refer to Algorithm 3 for its formal description. Subsequently, the online phase efficiently computes the approximate preimage by leveraging the semi-random sampling technique described in [25, 30]. That is, given $\mathbf{v} \in \mathcal{R}_Q^d$, sample $\mathbf{x} \in \mathcal{R}^{wd}$ such that $\mathbf{F} \cdot \mathbf{x} \approx \mathbf{v} \bmod Q$. It is easy to check that the convolution vector $\mathbf{y} = \mathbf{p} + \begin{bmatrix} \mathbf{R} \\ \mathbf{I} \end{bmatrix} \cdot \mathbf{x} \in \mathcal{R}^{(2+w)d}$ satisfies $\mathbf{A} \cdot \mathbf{y} \approx \mathsf{H}(id)$. Output the secret key $(\mathbf{y}_1, \ldots, \mathbf{y}_{w+1})$ by dropping the first component, since *mpk* is in the Hermite normal form (HNF), and we use the rANS encoding [31] to compress it.

4.2.1. Perturbation Sampling. On a high level, SampleP performs Genise and Micciancio's technique [32] on the module lattice. That is, the sampling process was carried out in a progressive manner, through iterative updates to the center and covariance matrix via the Schur complement. In more detail, it first samples $\mathbf{p}_2 \in \mathcal{R}^{wd}$ with variance $\sigma_2^2 - r^2$. Then sample $\mathbf{p}_1 \in \mathcal{R}^{2d}$ with center $\mathbf{c} = -\frac{r^2}{\sigma_2^2 - r^2} \cdot \mathbf{R} \cdot \mathbf{p}_2 \in \mathcal{K}_{\mathbb{R}}^{2d}$ and covariance $\Sigma_2 = \sigma_1^2 \cdot \mathbf{I}_{2d} - \frac{\sigma_2^2 \cdot r^2}{\sigma_2^2 - r^2} \mathbf{R}\mathbf{R}^t \in \mathcal{K}_{\mathbb{R}}^{2d \times 2d}$. To this end, it continues the above procedure recursively until it reaches the ring level, which is performed by calling SampleF in step 7. Notice that SampleF can repeat the above procedure as well by using the tower structures of $\mathcal{R}$ and $\mathcal{K}$ when the ring degree n is a power-of-2, as shown in [32]. In this work, we instead apply Peikert's Gaussian convolution method [23] at the ring level, thereby avoiding the need for sampling over $\mathbb{Z}$ with large and varying variances.

Algorithm 4 is simply a ring variant of Peikert's sampler [23]. In step 3, we abuse the notation, and it means that each coefficient c_i' of c' is used as input of SampleZ, that is, n independent parallel invocations of SampleZ, which sample integers with input c being the center, and fixed variance r^2. In this work, we use the method presented in [9], which samples from a fixed Gaussian distribution using a table-based method, after which a rejection step is applied to ensure output correctness. We omit the details.

4.2.2. Gadget Sampling. Given any $v \in \mathcal{R}_Q$, Algorithm 5 samples $\mathbf{x} \in \mathcal{R}^w$ such that $\mathbf{f} \cdot \mathbf{x} \approx v \bmod Q$, that is, $\mathbf{x}$ is an approximate image under the gadget $\mathbf{f} \in \mathcal{R}^w$. The procedure consists mainly of n parallel approximate gadget sampling ApproxGadget. In step 4, we get a vector $\mathbf{x} \in \mathcal{R}^w$ naturally from the w-by-n matrix formed by $\mathbf{z}_i$'s.

Algorithm 6 samples an approximate image $\mathbf{z}$ for input $u \in \mathbb{Z}_Q$ under the gadget vector $p \cdot \mathbf{g}$ over $\mathbb{Z}$ such that $\langle p \cdot \mathbf{g}, \mathbf{z} \rangle = u - e \bmod Q$ for a small error term $e \in \mathbb{Z}_p$, where e is deterministically generated in step 1 as described in [30]. It should be noted that step 3 can be implemented by invoking the

Table 3: Description of parameters and notations.

Parameters	Description
$\mathcal{B}^{32}$	Random seed space
(p, q)	Gadget parameters
Q	Modulus, $Q = pq$
b	A small integer as the log base
w	Dimension of gadget vector $\mathbf{f}$, $w = \lceil \log_b q \rceil$
n	A power of 2 integer, polynomial degree
$\mathcal{R}$	$\mathbb{Z}[X]/(X^n + 1)$
$\mathcal{R}_Q$	$\mathcal{R}/(Q \cdot \mathcal{R})$
d	Dimension of vectors or matrices over $\mathcal{R}$, $\mathcal{R}_Q$, or $\mathcal{K}_{\mathbb{R}}$
χ	A distribution over $\mathbb{Z}$
χ_n	A distribution over $\mathcal{R}$ where each entry follows χ
Q_s	Upper bound on the extraction query number
ε	Closeness parameter, $\varepsilon = 1/\sqrt{Q_s \cdot 256}$
r	Gaussian parameter $r = \eta_\varepsilon(\mathbb{Z}^{(2+w)\cdot(d+1)\cdot n})$
σ	Gadget Gaussian parameter $\sigma = br$
$\mathbf{f}$	Gadget vector $\mathbf{f} = p \cdot [1, b, \ldots, b^{w-1}] \in \mathcal{R}^w$
$\mathbf{g}$	Gadget vector $\mathbf{g} = [1, b, \ldots, b^{w-1}] \in \mathbb{Z}^w$
$\mathbf{R}$	Master secret key $\mathbf{R} \leftarrow \chi_n^{2d \times wd}$
$[\widehat{\mathbf{A}}, \mathbf{B}]$	Master public key, $\widehat{\mathbf{A}} \in \mathcal{R}_Q^{d \times d}$, $\mathbf{B} = \mathbf{F} - [\mathbf{I}_d, \widehat{\mathbf{A}}] \cdot \mathbf{R} \bmod Q$
$\mathbf{A}$	$\mathbf{A} = [\mathbf{I}_d, \widehat{\mathbf{A}}, \mathbf{B}]$, $\mathbf{A} \cdot \begin{bmatrix} \mathbf{R} \\ \mathbf{I} \end{bmatrix} = \mathbf{F} \bmod Q$
σ_1, σ_2	Standard deviation of the master secret key
Σ	Covariance for the *msk*, $\Sigma = \sigma_1^2 \cdot \mathbf{I} \oplus \sigma_2^2 \cdot \mathbf{I}$
$\Sigma_{\mathbf{p}}$	Covariance for the perturbation vectors $\Sigma_{\mathbf{p}} = \Sigma - \sigma^2 \cdot \begin{bmatrix} \mathbf{R} \\ \mathbf{I} \end{bmatrix} \cdot [\mathbf{R}^t \ \ \mathbf{I}] = \begin{bmatrix} \sigma_1^2 \cdot \mathbf{I} - \sigma^2 \mathbf{R}\mathbf{R}^t & -\sigma^2 \mathbf{R} \\ -\sigma^2 \mathbf{R}^t & (\sigma_2^2 - \sigma^2) \cdot \mathbf{I} \end{bmatrix} \succ r^2$
Σ_2	The Schur complement of $(\sigma_2^2 - \sigma^2) \cdot \mathbf{I}$ in $\Sigma_{\mathbf{p}}$, i.e., $\Sigma_2 = \sigma_1^2 \cdot \mathbf{I} - \frac{\sigma_2^2 \cdot \sigma^2}{\sigma_2^2 - \sigma^2} \mathbf{R}\mathbf{R}^t$
H	Hash function $\mathsf{H}: \{0, 1\}^* \to \mathcal{R}_Q^d$
$\mathbf{y}$	User's secret key, $\mathbf{A} \cdot \mathbf{y} = \mathsf{H}(id) \bmod Q$
ζ	$\zeta = X^{n/2} + 1$, a polynomial such that $\zeta\zeta^* = 2 \bmod X^n + 1$
$\mathbf{j}$	$\mathbf{j} = (\zeta^*, 0, \ldots, 0) \in \mathcal{R}^d$
$\mathbf{b}$	Public key derived from user's *id*, $\mathbf{b} = Q \cdot \mathbf{j} - 2\mathsf{H}(id) \bmod 2Q$
$\bar{\mathbf{A}}$	$\bar{\mathbf{A}} = [2\mathbf{I}_d, 2\widehat{\mathbf{A}}, 2\mathbf{B}, \mathbf{b}] = [2\mathbf{A}, \mathbf{b}] \in \mathcal{R}_Q^{d \times ((2+w)d+1)}$, $\bar{\mathbf{A}} \cdot \begin{bmatrix} \mathbf{y} \\ 1 \end{bmatrix} = Q \cdot \mathbf{j} \bmod 2Q$
γ	The upper-bound on the ℓ_2-norm of signatures
$\bar{\sigma}_1, \bar{\sigma}_2$	Standard deviation of the signature
$\bar{\Sigma}$	Covariance for the signatures, $\bar{\Sigma} = \bar{\sigma}_1^2 \cdot \mathbf{I}_{2d} \oplus \bar{\sigma}_2^2 \cdot \mathbf{I}_{wd+1}$
$\bar{\Sigma}_{\mathbf{p}}$	Covariance for the perturbation vectors $\bar{\Sigma}_{\mathbf{p}} = \bar{\Sigma} - r^2 \cdot \begin{bmatrix} \mathbf{y} \\ 1 \end{bmatrix} \cdot [\mathbf{y}^t \ \ 1] \succ r^2$
$\bar{\mathsf{H}}$	Another hash function $\bar{\mathsf{H}}: \{0, 1\}^* \to \{0, 1\}^n$

SampleZ function: each z_i is computed as $z_i \leftarrow b \cdot \mathsf{SampleZ}(-v/b) + v$.

4.3. Signing. Given *mpk*, *id*, sk_{id} and a message $\mu \in \{0, 1\}^*$, IBS.Sign perform the G + G signature [26] essentially, which, similar to IBS.Extract, divides into offline/online phases, as they share the same underlying techniques. Refer to Algorithm 7 for more details.

4.4. Verification. On input *mpk*, *id*, μ and its signature *sig*, the verification Algorithm 8 checks that the signature $\mathbf{z}$ is short and verifies $c = \bar{\mathsf{H}}(\mathbf{A}\mathbf{z} - Q \cdot c \cdot \mathbf{j} \bmod 2Q, \mu)$.

Now we analyze the correctness of the scheme. By definition, user's secret key $\mathbf{y}$ satisfies $[\mathbf{I}_d, \widehat{\mathbf{A}}, \mathbf{B}] \cdot \mathbf{y} = \mathsf{H}(id) \bmod Q$, that is $\bar{\mathbf{A}} \cdot \begin{bmatrix} \mathbf{y} \\ 1 \end{bmatrix} = Q \cdot c \cdot \mathbf{j} \bmod 2Q$. Besides, the output signature $\mathbf{z} \sim$

Ensure: $(mpk,\ msk)$
1: $\mathsf{seed}_{\widehat{\mathbf{A}}} \leftarrow \mathcal{B}^{32}$
2: $\widehat{\mathbf{A}} \leftarrow \mathsf{XOF}(\mathsf{seed}_{\widehat{\mathbf{A}}})$ ▷ $\widehat{\mathbf{A}} \in \mathcal{R}_Q^{d\times d}$
3: **repeat**
4: $\mathbf{R} \leftarrow \chi_n^{2d\times wd}$ ▷ $\mathbf{R} \in \mathcal{R}^{2d\times wd}$
5: $\Sigma_{\mathbf{p}} = \Sigma - \sigma^2 \cdot \begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix} \cdot [\mathbf{R}^t\ \mathbf{I}]$
6: **until** $\Sigma_{\mathbf{p}} > r^2$
7: $\mathbf{F} = \mathbf{I}_d \otimes p \cdot [1,\ b,\ldots,\ b^{w-1}]$ ▷ $\mathbf{f} = p \cdot [1,\ b,\ldots,\ b^{w-1}] \in \mathcal{R}^{1\times w}$
8: $\mathbf{B} = \mathbf{F} - [\mathbf{I}_d,\ \widehat{\mathbf{A}}] \cdot \mathbf{R} \bmod Q$ ▷ $\mathbf{B} \in \mathcal{R}_Q^{d\times wd}$
9: **return** $(mpk = (\mathsf{seed}_{\widehat{\mathbf{A}}}, \mathbf{B}),\ msk = \mathbf{R})$

Algorithm 1: IBS.Setup.

Require: msk,$id \in \{0,\ 1\}^*$
Ensure: sk_{id}
Offline phase:
1: $\mathbf{p} \leftarrow \mathsf{SampleP}(\mathbf{R})$ ▷ $\mathbf{p} \in \mathcal{R}^{(2+w)d}$
2: $\widehat{\mathbf{A}} \leftarrow \mathsf{XOF}(\mathsf{seed}_{\widehat{\mathbf{A}}})$ ▷ $\widehat{\mathbf{A}} \in \mathcal{R}_Q^{d\times d}$
3: $\mathbf{A} = [\mathbf{I}_d,\ \widehat{\mathbf{A}},\ \mathbf{B}]$ ▷ $\mathbf{A} \in \mathcal{R}_Q^{d\times(2+w)d}$
4: $\mathbf{v} = \mathsf{H}(id) - \mathbf{A} \cdot \mathbf{p} \bmod Q$ ▷ $\mathbf{v} = (v_0, \cdots, v_{d-1}) \in \mathcal{R}_Q^d$
Online phase:
5: **for** $i = 0$ to $d-1$ **do**
6: $\mathbf{x}_i \leftarrow \mathrm{SampleGadgetF}(v_i)$ ▷ $\mathbf{x}_i \in \mathcal{R}^w,\ \langle \mathbf{f}, \mathbf{x}_i\rangle \approx v_i \bmod Q$
7: **end for**
8: $\mathbf{x} = (\mathbf{x}_0, \ldots, \mathbf{x}_{d-1})$ ▷ $\mathbf{x} \in \mathcal{R}^{wd}$
9: $\mathbf{y} = \mathbf{p} + \begin{bmatrix}\mathbf{R}\\ \mathbf{I}\end{bmatrix} \cdot \mathbf{x} \in \mathcal{R}^{(2+w)d}$ ▷ $\mathbf{y} = (\mathbf{y}_0, \ldots, \mathbf{y}_{w+1})$
10: $\mathbf{y}' = (\mathbf{y}_1, \ldots, \mathbf{y}_{w+1}) \in \mathcal{R}^{(1+w)d}$ ▷ $\left[\widehat{\mathbf{A}}, \mathbf{B}\right] \cdot \mathbf{y}' \approx \mathsf{H}(id) \bmod Q$
11: **return** $sk_{id} \leftarrow \mathbf{y}'$

Algorithm 2: IBS.Extract (mpk, id).

$D_{\mathcal{R}^{(2+w)d+1},\tilde{\Sigma}}$ is a short vector, and satisfies the following:

$$\begin{aligned}\mathbf{A}\cdot\mathbf{z} &= \mathbf{A}\cdot\left(\mathbf{p} + \begin{bmatrix}\mathbf{y}\\ 1\end{bmatrix}(\zeta\cdot u + c)\right)\\ &= \mathbf{w} + Q\cdot c\cdot\mathbf{j}\cdot(\zeta\cdot u + c)\\ &= \mathbf{w} + Q\cdot c\cdot\mathbf{j} \bmod 2Q.\end{aligned}$$

Therefore, the user with the secret key can produce a valid signature.

4.5. Security in the QROM. Following [33], SRGG-IBS can be regard as an ordinary signature scheme $\mathsf{SIG} = (\mathsf{SIG.Gen},$

Require: msk
Ensure: $\mathbf{p} \sim D_{\mathcal{R}^{(2+w)d},\sqrt{\Sigma_{\mathbf{p}}}}$
1: $\mathbf{p}_2 \leftarrow D_{\mathbb{Z}^{wd\cdot n},\sqrt{\sigma_2^2-\sigma^2}}$ ▷ $\mathbf{p}_2 \sim D_{\mathcal{R}^{wd},\sqrt{\sigma_2^2-\sigma^2}}$
2: $\mathbf{c} = -\frac{\sigma^2}{\sigma_2^2-\sigma^2}\cdot\mathbf{R}\cdot\mathbf{p}_2$ ▷ $\mathbf{c} \in \mathcal{K}_{\mathbb{R}}^{2d}$
3: $\Sigma_2 = \sigma_1^2\cdot\mathbf{I}_{2d} - (\sigma^{-2} - \sigma_2^{-2})^{-1}\mathbf{R}\mathbf{R}^T$ ▷ $\Sigma \in \mathcal{K}_{\mathbb{R}}^{2d\times 2d}$
4: **for** $i = 2d-1$ to 0 **do**
5: $\mathbf{c} = (\mathbf{c}', c_i)$ ▷ $\mathbf{c}' \in \mathcal{K}_{\mathbb{R}}^{i}, c_i \in \mathcal{K}_{\mathbb{R}}$
6: $\Sigma_2 = \begin{bmatrix}\mathbf{A} & \mathbf{B}\\ \mathbf{B}^T & f\end{bmatrix}$ ▷ $\mathbf{A} \in \mathcal{K}_{\mathbb{R}}^{i\times i}, \mathbf{B} \in \mathcal{K}_{\mathbb{R}}^{i\times 1}, f \in \mathcal{K}_{\mathbb{R}}$
7: $p_i \leftarrow \mathsf{SampleF}(f, c_i)$ ▷ $p_i \in \mathcal{R}$
8: $\mathbf{c} = \mathbf{c}' + f^{-1}\mathbf{B}(p_i - c_i)$ ▷ $\mathbf{c} \in \mathcal{K}_{\mathbb{R}}^{i}$
9: $\Sigma_2 = \mathbf{A} - f^{-1}\mathbf{B}\mathbf{B}^T$ ▷ $\Sigma_2 \in \mathcal{K}_{\mathbb{R}}^{i\times i}$
10: **end for**
11: $\mathbf{p}_1 \leftarrow (p_0, \ldots, p_{2d-1})$ ▷ $\mathbf{p}_1 \in \mathcal{R}^{2d}$
12: **return** $\mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2)$ ▷ $\mathbf{p} \in \mathcal{R}^{(2+w)d}$

Algorithm 3: SampleP $(\mathbf{R})$.

Require: $f, c \in \mathcal{K}_{\mathbb{R}}$
Ensure: $x \leftarrow D_{\mathcal{R},f,c}$
1: $y \leftarrow \mathcal{N}_n$
2: $c' = c + \sqrt{d - r^2}\cdot y$ ▷ $c' = c_0' + \cdots + c_{n-1}'\cdot X^{n-1} \in \mathcal{K}_{\mathbb{R}}$, $r \geq \eta_\varepsilon(\mathcal{R}^2)$
3: $x \leftarrow \mathsf{SampleZ}(c')$ ▷ $x \in \mathcal{R}$
4: **return** x

Algorithm 4: SampleF.

Require: $v \in \mathcal{R}_Q$ ▷ $v(X) = v_0 + v_1X + \cdots + v_{n-1}X^{n-1}$
Ensure: $\mathbf{x} \in \mathcal{R}^w$
1: **for** $i = 0$ to $n-1$ **do**
2: $\mathbf{z}_i \leftarrow \mathsf{ApproxGadget}(v_i)$ ▷ $\mathbf{z}_i \in \mathbb{Z}^w$
3: **end for**
4: $[\mathbf{z}_0, \ldots,\ \mathbf{z}_{n-1}] \Rightarrow (x_0, \ldots, x_{w-1}) = \mathbf{x} \in \mathcal{R}^w$
5: **return** $\mathbf{x}$ ▷ $\mathbf{f}\cdot\mathbf{x} \approx v \bmod Q$

Algorithm 5: SampleGadgetF.

Require: $u \in \mathbb{Z}_Q$
Ensure: $\mathbf{z} \in \mathbb{Z}^w$ ▷ $\langle p\cdot\mathbf{g}, \mathbf{z}\rangle \approx u \bmod Q$
1: $e = u \bmod p,\ v = (u-e)/p$
2: **for** $i = 0$ to $w-1$ **do**
3: $z_i \leftarrow D_{b\mathbb{Z}+v,r}$
4: $v = \frac{v-z_i}{b}$
5: **end for**
6: **return** $\mathbf{z} = (z_1, \ldots, z_w)$

Algorithm 6: ApproxGadget.

Require: mpk, id, sk_{id}, μ
Ensure: sig
Offline phase:
$\mathbf{y}_0 = \mathsf{H}(id) - \left[\widehat{\mathbf{A}},\ \mathbf{B}\right] \cdot \mathbf{y}' \bmod Q$ ▷ $\mathbf{y}_0 \in \mathscr{R}^d$
$\mathbf{y} = (\mathbf{y}_0, \mathbf{y}')$ ▷ $\mathbf{y} \in \mathscr{R}^{(2+w)d}$
$\mathbf{p} \leftarrow \mathsf{SampleP}(\mathbf{y})$ ▷ $\mathbf{p} \sim D_{\mathscr{R}^{(2+w)d+1}, \Sigma_{\mathbf{p}}}$
$\widehat{\mathbf{A}} \leftarrow \mathsf{XOF}(\mathsf{seed}_{\widehat{\mathbf{A}}})$ ▷ $\widehat{\mathbf{A}} \in \mathscr{R}_Q^{d \times d}$
$\mathbf{b} = Q \cdot \mathbf{j} - 2\mathsf{H}(id) \bmod 2Q$ ▷ $\mathbf{b} \in \mathscr{R}^d$
$\bar{\mathbf{A}} = [2\mathbf{I}_d,\ 2\widehat{\mathbf{A}},\ 2\mathbf{B},\ \mathbf{b}]$ ▷ $\bar{\mathbf{A}} \in \mathscr{R}_{2Q}^{d \times ((2+w)d+1)}$
$\mathbf{w} = \bar{\mathbf{A}} \cdot \mathbf{p} \bmod 2Q$
Online phase:
$c \leftarrow \bar{\mathsf{H}}(\mathbf{w}, \mu)$ ▷ $c \in \{0, 1\}^n$
$u \leftarrow D_{\mathscr{R}, r, -c/\zeta}$
$\mathbf{z} = \mathbf{p} + \begin{bmatrix} \mathbf{y} \\ 1 \end{bmatrix} \cdot (\zeta u + c)$ ▷ $\mathbf{z} \in \mathscr{R}^{(2+w)d+1}$
if $\|\mathbf{z}\| > \gamma$ **then**
    restart
end if
return $sig = (\mathbf{z}, c)$

Algorithm 7: IBS.Sign (mpk, id, sk_{id}, μ).

Require: mpk, id, sig
Ensure: accept or reject
$\widehat{\mathbf{A}} \leftarrow \mathsf{XOF}(\mathsf{seed}_{\widehat{\mathbf{A}}})$ ▷ $\widehat{\mathbf{A}} \in \mathscr{R}_Q^{d \times d}$
$\bar{\mathbf{A}} = [2\mathbf{I}_d,\ 2\widehat{\mathbf{A}},\ 2\mathbf{B},\ \mathbf{b}]$ ▷ $\bar{\mathbf{A}} \in \mathscr{R}_{2Q}^{d \times ((2+w)d+1)}$
$\mathbf{w} = \mathbf{A}\mathbf{z} - Q \cdot c \cdot \mathbf{j} \bmod 2Q$ ▷ $\mathbf{w} \in \mathscr{R}^d$
if $c = \bar{\mathsf{H}}(\mathbf{w}, \mu)$ and $\|\mathbf{z}\| \le \gamma$ **then**
    return accept
end if
return reject

Algorithm 8: IBS.Vrfy(mpk, id, μ, sig).

SIG.Sign, SIG.Vrfy), but there exists a trapdoor such that the secret key can be derived from the public key, meaning that $\mathsf{SIG.Gen}(1^n)$ can be realized by using the preimage sampleable functions (PSFs) [16] as follows: $(a, t) \leftarrow \mathsf{PSF.TrapGen}(1^n)$, sample $sk \leftarrow \mathsf{PSF.SampleDom}(1^n)$, compute $f_a(sk)$ and output $(pk, sk) = ((a, f_a(sk)), sk)$. Then SRGG-IBS is described alternatively as follows:

- $\mathsf{IBS.Setup}(1^\lambda)$: $(a, t) \leftarrow \mathsf{PSF.TrapGen}(1^\lambda)$ and output $(mpk, msk) = (a, t)$.
- $\mathsf{IBS.Extract}(mpk, msk, id)$: the extraction procedure simply runs $sk_{id} \leftarrow \mathsf{PSF.SamplePre}(t, \mathsf{H}(id))$.
- $\mathsf{IBS.Sign}(mpk, sk_{id}, id, \mu)$: the signing procedure runs $sig \leftarrow \mathsf{SIG.Sig}(sk_{id}, \mathsf{H}(id), \mu)$.
- $\mathsf{IBS.Vrfy}(mpk, id, sig)$: output $\mathsf{SIG.Vrfy}(id, \mu, sig)$.

Theorem 2. *Let* IBS *and* SIG *are constructed as above. If* SIG *is sEUF-CMA secure, then* IBS *is sEUF-ID-CMA secure in the QROM.*

Proof. Let $\mathscr{A}$ represent a quantum adversary that can issue at most q_H queries to the hash oracle $\mathscr{O}_H$, q_S queries to the signing oracle $\mathscr{O}_{\text{sign}}$, and q_E queries to the key extraction oracle $\mathscr{O}_{\text{extract}}$, and that breaks IBS with nonnegligible advantage ε.

Game G_0: Game G_0 denotes the standard attack game for IBS. The challenger produces (mpk, msk) from IBS.Setup, and sends mpk to $\mathscr{A}$. Then, the adversary $\mathscr{A}$ is permitted to issue classical extraction queries on identities id_i, and the challenger replies to each query by returning $sk_{id} \leftarrow \mathsf{IBS.Extract}(mpk, msk, id_i)$ and save (id, sk_{id}) in list $\mathscr{L}_{id}$. If $\mathscr{A}$ queries $\mathscr{O}_{\text{sign}}$ on (id, μ), then the challenger returns $sig \leftarrow \mathsf{IBS.Sign}(mpk, sk_{id}, id, \mu)$ and saves (id, μ, sig) in $\mathscr{L}_\mu$. Additionally, the adversary $\mathscr{A}$ may perform quantum queries to the random oracle H. Finally, $\mathscr{A}$ generates an identity (id^*, μ^*, sig^*). If $(id^*, \cdot) \notin \mathscr{L}_{id}, (id^*, \mu^*, \cdot) \notin \mathscr{L}_\mu$, and $\mathsf{IBS.Vrfy}(mpk, id^*, \mu^*, sig^*) = 1$, then $\mathscr{A}$ wins the game, an event that occurs with probability ε by definition.

Game G_1: Let $\delta \in (0, 1)$. Game G_1 is defined as follows: First pick a subset $\mathscr{X} \subset \mathscr{FD}$ such that $Pr[id \in \mathscr{X}] = \delta$. Then the conditions for a successful attack are changed to add further restrictions to the original ones: $id^* \in \mathscr{X}$ and $id_i \notin \mathscr{X}$ for each i. Since $\mathscr{A}$ never queries id^*'s secret key and each identity happens in $\mathscr{X}$ has independent probability δ, then $\mathscr{A}$ has success probability at least

$$\delta(1 - \delta \cdot q_S) \cdot \varepsilon \ge \delta \cdot \varepsilon - q_S \cdot \delta^2.$$

Game G_2: Game G_2 is the same as G_1, except that we redefine the hash function: sample a random pk for SIG, and define $H(x) = pk$ for all $x \in \mathscr{X}$, while assigning $H(x)$ uniformly at random for all other inputs. The function H is then distributed according to the semi-constant distribution SC_δ. By an (adapted) claim in [33], there exists a polynomial $l(\cdot, \cdot, \cdot)$ for which adversary $\mathscr{A}$ succeeds in game G_2 with probability no less than

$$\delta \cdot \varepsilon - \frac{l(q_H, q_S, q_E)^2}{\delta}.$$

Now, we construct a PPT adversary $\mathscr{B}$ breaking SIG. Suppose that algorithm $\mathscr{B}$ possesses quantum query access to two random oracles, denoted $\mathscr{O}_1$ and $\mathscr{O}_2$. Here, $\mathscr{O}_1$ maps identity strings to the randomness required by PSF.SampleDom, while $\mathscr{O}_2$ maps identities to binary values, returning 1 with probability δ. Given the input (mpk, msk), the procedure of $\mathscr{B}$ is described as follows:

- Send the master public key mpk to adversary $\mathscr{A}$ and emulate its execution, while acting as the challenger in the interaction with $\mathscr{A}$.
- Construct a quantum random oracle H with the property that

$$H(id) = \begin{cases} pk & \text{if } \mathrm{O}_2(id) = 1 \\ f_{\mathrm{mpk}}(\mathsf{FPS.SampleDom}(1^{\lambda}; \mathrm{O}_1(id))) & \text{else} \end{cases}.$$

- When adversary $\mathscr{A}$ requests the secret key corresponding to an identity id_i, the algorithm evaluates $\mathrm{O}_2(id_i)$. If the output equals 1, the process terminates by returning a random bit. If not, the response is generated by $sk_i \leftarrow \mathsf{FPS.SampleDom}(1^{\lambda}; \mathrm{O}_1(id))$.
- When $\mathscr{A}$ makes a signing query for (id, μ), if $\mathscr{A}$ never asks the secret key query for id, then $\mathscr{B}$ computes $\mathrm{O}_2(id)$. If the outcome is 1, the algorithm outputs a randomly selected bit and terminates immediately. Otherwise, compute $sk_{id_i} \leftarrow \mathsf{FPS.SampleDom}(1^n; \mathrm{O}_1(id))$ and returns $sig \leftarrow \mathsf{IBS.Sign}(mpk, sk_{id}, id, \mu)$.
- When $\mathscr{A}$ outputs (id^*, μ^*, sig^*), $\mathscr{B}$ outputs (sig^*, μ^*).

Let $\mathscr{X} = \{id \in \mathscr{ID} | \mathrm{O}_2(id) = 1\}$. We can see that the abort conditions are equivalent to game G_2. For each extraction query id_i that succeeds, we have $\mathrm{O}_2(id_i) = 0$, so

$$f_{mpk}(sk_{id_i}) = f_{mpk}(\mathsf{PSF.SampleDom}(1^n; \mathrm{O}_1(id_i))) = H(id_i).$$

Therefore, sk_i constitutes a valid secret key for identity id_i and is properly distributed. Provided that $\mathscr{B}$ does not abort during the challenge query, this implies $\mathrm{O}_2(id^*) = 1$, and consequently $H(id^*) = pk$. Note that H is distributed according to the semi-constant distribution SC_{δ}. Therefore, if $\mathscr{B}$ does not abort, we win if and only if $\mathscr{A}$ wins. Thus, the view of $\mathscr{A}$ as a subroutine of $\mathscr{B}$ is the same as in game G_2. Therefore, $\mathscr{B}$ wins with probability at least $\delta \cdot \varepsilon - \frac{l(q_H, q_S, q_E)}{4}\delta^2$. The maximum advantage occures when $\delta = 22\varepsilon / l(q_H, q_S, q_E)$, which gives success probability at least $\frac{\varepsilon^2}{l(q_H, q_S, q_E)}$. Thus, if $\mathscr{A}$ has nonnegligible probability, so does $\mathscr{B}$. □

5. Specification and Implementation

5.1. Recommended Parameters. The basic principle of our parameter selection is to balance the security and flexibility of the instantiation. We provide three parameter sets in Table 4 for target security 100, 180, and 256. In all parameter sets, we fix the polynomial degree $n = 512$, gadget parameter $(p, q) = (2^{14} - 1, 31^2)$, and log base $b = 31$. Thus, the modulus $Q = p \cdot q = 15744063$ and the gadget dimension $w = \lceil \log_b q \rceil = 2$ are fixed as well. Besides, the distribution χ is fixed to the centered binomial distribution with parameter 1, that is, $\{-1 : \frac{1}{4}, 0 : \frac{1}{2}, 1 : \frac{1}{4}\}$, thus χ_n denotes the centered binomial distribution over $\mathscr{R}$. Finally, we set the upper bound of the extraction query number $Q_s = 2^{40}$. As a result, the base Gaussian parameter $r = \eta_{\varepsilon}(\mathbb{Z}^{(2+w)\cdot 5\cdot 512}) \approx 1.16$, where $\varepsilon = 1/\sqrt{Q_s \cdot 256}$.

In Table 4, the estimations for the secret key size and signature size are made according to the entropy of the preimage. These values can be efficiently achieved through batch encoding via the asymmetric numeral system (ANS), following the approach described in [31].

5.2. Concrete Security. The concrete security level is assessed using standard cryptanalytic approaches established for lattice-based cryptography. Specifically, we evaluate the computational cost associated with the most effective attacks targeting master secret key recovery and signature forgery. This cost is then converted into concrete bit-security estimates within the Core-SVP framework [34]. It should be noted that although our construction is based on the MLWE and MSIS problems over polynomial rings, no known attacks currently exploit this ring-specific structure. As a result, the most powerful attacks treat these problems generically as standard LWE and SIS instances.

5.2.1. Lattice Reduction and the Core-SVP Model. Lattice reduction involves the computation of a basis composed of short and nearly orthogonal vectors, serving as a fundamental technique in cryptanalysis for attacking lattice-based systems. The most widely used algorithms in practice are BKZ [35] and its enhanced versions [36, 37]. The BKZ algorithm is configured by the blocksize parameter β. Given a lattice Λ of dimension d, BKZ-β produces a nonzero vector $\mathbf{v} \in \Lambda$ satisfying

$$\|\mathbf{v}\| \leq \delta_{\beta}^{d} \mathsf{vol}(\Lambda)^{1/d} \text{ and } \delta_{\beta} \approx \left(\frac{(\pi\beta)^{\frac{1}{\beta}}\beta}{2\pi e}\right)^{\frac{1}{2(\beta-1)}},$$

for $\beta > 50$. In the Core-SVP model [34], the computational cost of executing BKZ-β is approximated as $2^{0.292\beta}$ classically and $2^{0.265\beta}$ under quantum settings. This model provides a conservative and widely accepted measure for concrete bit-security.

5.2.2. Master Secret Key Recovery Attack. Recovering the master secret key in our scheme requires finding a short secret pair $(\mathbf{s}, \mathbf{e})$ satisfying the equation $\mathbf{As} + \mathbf{e} = \mathbf{b} \bmod Q$, where $\mathbf{A} \in \mathbb{Z}_Q^{m \times m}$ and $\mathbf{b} \in \mathbb{Z}_Q^m$ are public. A primary approach for this task is the primal attack. The attack executes BKZ-β on the lattice $\Lambda = \Lambda_Q^{\perp}([\mathbf{A}|\mathbf{I}_n|\mathbf{b}])$ of dimension $d = 2m + 1$, aiming to recover the short vector $(\mathbf{s}, \mathbf{e}, -1)$. As established in [34], successful key recovery is achievable when the block size β meets the condition

$$\|(\mathbf{s}, \mathbf{e}, -1)\| \sqrt{\frac{\beta}{d}} \leq \delta_{\beta}^{2\beta - d - 1} \cdot Q^{\frac{m}{d}}.$$

5.2.3. Forgery Attack. The signature forgery in our schemes is essentially to solve an approximate-CVP instance over the Q-ary lattice. The nearest-colattice algorithm [38] is a primary approximate-CVP algorithm. Specifically, given $(\mathbf{A}, \mathbf{u}) \in \mathbb{Z}_Q^{m \times d} \times \mathbb{Z}_Q^n$ and $\beta > 0$, a preimage $\mathbf{x} \in \mathbb{Z}^n \times \mathbb{Z}^{d-n}$ such that $\mathbf{Ax} = \mathbf{u} \bmod Q$ can be computed by BKZ with blocksize β satisfying as follows:

Table 4: Parameters and sizes (in bytes).

Parameters	PARAM-I	PARAM-II	PARAM-III
Target security	100	180	256
Module rank d	2	3	4
Std. dev. of msk σ_1	2843.9	3399.4	3867.7
Std. dev. of msk σ_2	1023.0	1065.3	1089.9
Std. dev. of sig $\bar{\sigma}_1$	843, 088.2	1, 046, 714.2	1, 238, 066.0
Std. dev. of sig $\bar{\sigma}_2$	310, 891.6	384, 614.5	448, 859.2
Acceptance bound γ	42, 086, 491.7	63, 653, 172.0	86, 573, 930.5
Master public key size $\|mpk\|$	12,320	27,680	49,184
Master secret key size $\|msk\|$	2048	4608	8192
Secret key size $\|sk_{id}\|$	4900	7396	9892
Signature size $\|sig\|$	12,148	17,786	23,516

Table 5: Concrete security.

Security	PARAM-I	PARAM-II	PARAM-III
Blocksize for key recov.	358	608	874
Classical core-SVP	105	178	255
Quantum core-SVP	95	161	232
Blocksize for forg. attack	352	586	828
Classical core-SVP	103	171	242
Quantum core-SVP	93	155	219

Table 6: Performance numbers are the median cycle measured over 1000 executions.

Algorithms	PARAM-I	PARAM-II	PARAM-III
Setup	1,520,313	3,445,003	7,294,250
Extract	5,004,676	5,490,761	9,556,676
Sign	5,203,248	7,007,089	11,521,424
Vrfy	1,218,478	1,762,221	2,544,205

$$\beta \geq \min_{k \leq d-m} \left(\delta_\beta^{d-k} Q^{\frac{m}{d-k}} \right).$$

The security estimations are listed in Table 5.

5.3. Performance. To evaluate the practical performance of our construction, we developed an initial implementation written entirely in C. This implementation was compiled using gcc 8.3.0 and executed on a system running Deepin 20.9. Performance metrics, measured on a single core of an Intel Core i7-10710U processor operating at 1.1 GHz with 8 GB of RAM, are summarized in Table 6.

To better visualize the efficiency of our algorithms, we list in Table 7 the number of times each algorithm is run in 1 s. Our implementations will be publicly available soon.

Our proof-of-concept implementation involves a large number of floating-point operations. However, it can be easily implemented fully over integers with the technique of [39] in both the key extraction and signing procedures, which will make SRGG-IBS an attractive post-quantum IBS for constrained environments. In addition, we may use the techniques in [40] that support NTT multiplication for our NTT-unfriendly modulus Q. We defer the development of an optimized implementation and the integration of provable side-channel countermeasures to future work.

Table 7: Performance number of times each algorithm is run in 1 s.

Algorithms	PARAM-I	PARAM-II	PARAM-III
Setup	222	62	27
Extract	104	46	25
Sign	88	42	26
Vrfy	236	87	48

6. Conclusion

We propose an efficient IBS scheme constructed on ideal lattices, designed for practical applications. Our approach follows the "preimage sampling [25] + Ellipsoid G + G" framework and provides security proofs in the QROM. Furthermore, we offer three distinct parameter sets and a proof-of-concept implementation. Owing to the compact gadget framework introduced in [25], the key extraction process is both efficient and straightforward, while also maintaining small key sizes. These advantages position SRGG-IBS as a promising post-quantum IBS candidate suitable for resource-constrained environments. We believe the combination of the preimage sampling with the Fiat–Shamir transformation may be useful for the practical instantiation of other advanced signature schemes.

Data Availability Statement

The data that support the findings of this study are available upon request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

Conflicts of Interest

The authors declare no conflicts of interest.

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Endnotes

[1]Note that the parameters suggested in [21] are not very informative, as their security estimations are far from the current ones.

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